Step 1 · Build and solve the decay model

Let A(t) be the mass in milligrams, with t measured in hours.

$$\frac{dA}{dt} = kA, \qquad A(0) = 100, \qquad k < 0$$

$$\frac{dA}{A} = k\,dt \quad \Longrightarrow \quad \ln|A| = kt + C$$

$$A(t) = C_1 e^{kt}, \qquad A(0) = 100 \Longrightarrow C_1 = 100$$

$$A(t) = 100e^{kt}$$

Problem 21

A dead body was found within a closed room of a house where the temperature was a constant 70°F. At the time of discovery, the core temperature of the body was determined to be 85°F. One hour later a second measurement showed that the core temperature of the body was 80°F.

Assume that the time of death corresponds to $t = 0$ and that the core temperature at that time was 98.6°F. Determine how many hours elapsed before the body was found.

Step 1 · Model the temperature difference

Newton's law of cooling uses the difference from the room temperature.

$$\frac{dT}{dt} = k(T - 70), \qquad k < 0$$

$$\frac{dT}{T - 70} = k\,dt \implies \ln|T - 70| = kt + C$$

$$T(t) = 70 + C_1 e^{kt}, \qquad T(0) = 98.6 \implies C_1 = 28.6$$

$$T(t) = 70 + 28.6e^{kt}$$

Step 2 · Use both temperature measurements

Let n be the number of hours from death to discovery. The second reading is at $n + 1$.

$$T(n) = 85 \implies 28.6e^{kn} = 15$$

$$T(n+1) = 80 \implies 28.6e^{k(n+1)} = 10$$

$$\frac{28.6e^{k(n+1)}}{28.6e^{kn}} = \frac{10}{15} \implies e^{k} = \frac{2}{3}$$

$$k = \ln\left(\frac{2}{3}\right) \approx -0.405465\ \text{h}^{-1}$$

Take the ratio of the excess temperatures: 10/15, not 80/85.

Step 3 · Solve for the time before discovery

Return to the first observation and substitute the value of k.

$$28.6e^{kn} = 15 \quad \Longrightarrow \quad e^{kn} = \frac{15}{28.6}$$

$$kn = \ln\left(\frac{15}{28.6}\right)$$

$$n = \frac{\ln(15/28.6)}{\ln(2/3)} \approx 1.59165 \text{ hours} \approx 1.6 \text{ hours}$$

This is the time from death to discovery. The second temperature reading was taken one hour after discovery.

Problem 31

A 30-volt electromotive force is applied to an LR-series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms.

Find the current $i(t)$ if $i(0) = 0$. Determine the current as $t \to \infty$.

Step 2 · Apply the initial condition and take the limit

$$i(0) = 0 \quad \Longrightarrow \quad 0 = 0.6 + C \quad \Longrightarrow \quad C = -0.6$$

$$i(t) = 0.6\left(1 - e^{-500t}\right) \text{ A}$$

As time increases, the exponential transient tends to zero.

$$\lim_{t \to \infty} i(t) = 0.6 \text{ A}$$

Checks: $i(0) = 0$, and the steady current is $E/R = 30/50 = 0.6$ A.

Step 1 · Write the circuit equation and integrate

$$L\frac{di}{dt} + Ri = E(t) \quad \Longrightarrow \quad 0.1\frac{di}{dt} + 50i = 30$$

$$\frac{di}{dt} + 500i = 300, \qquad \mu(t) = e^{500t}$$

$$\frac{d}{dt}\left(e^{500t} i\right) = 300e^{500t}$$

$$e^{500t} i = \frac{300}{500} e^{500t} + C \quad \Longrightarrow \quad i(t) = 0.6 + Ce^{-500t}$$

Problem 32

Solve equation (8) under the assumption that $E(t) = E_0 \sin(\omega t)$ and $i(0) = i_0$.

Equation (8): $$L\frac{di}{dt} + Ri = E(t)$$

$$E(t) = E_0 \sin(\omega t), \qquad i(0) = i_0$$

Step 1 · Use an integrating factor

Write a = R/L to keep the integration readable.

$$\frac{di}{dt} + ai = \frac{E_0}{L}\sin(\omega t), \qquad a = \frac{R}{L}, \qquad \mu(t) = e^{at}$$

$$\frac{d}{dt}\left(e^{at} i\right) = \frac{E_0}{L} e^{at} \sin(\omega t)$$

$$e^{at} i = \frac{E_0}{L} J + C, \qquad J = \int e^{at} \sin(\omega t)\, dt$$

The remaining task is to integrate an exponential multiplied by a sine.

$$\int \frac{E_0}{L} e^{at} \sin(\omega t) = \frac{E_0}{L}\left[\frac{1}{a} e^{at} \sin(\omega t) - \int \frac{\omega}{a} e^{at} \cos(\omega t)\, dt\right]$$

$$\frac{E_0}{L}\left[\frac{1}{a} e^{at} \sin(\omega t) - \frac{\omega}{a} \int e^{at} \cos(\omega t)\, dt\right]$$

B

$dv = e^{at} \Rightarrow v = \frac{1}{a} e^{at}$

$u = \cos(\omega t)$ $\quad du = -\omega \sin(\omega t)$

$$B = \frac{1}{a} e^{at} \cos(\omega t) - \int \frac{\omega}{a} e^{at} \cdot \sin(\omega t)$$

I

$$I = \frac{1}{a}\left[e^{at}\sin(\omega t) - \omega\left(\frac{1}{a}e^{at}\cos(\omega t) - I\right)\right]$$

$$I = \frac{e^{at}\sin(\omega t)}{a} - \frac{\omega}{a}e^{at}\cos(\omega t) - \frac{\omega}{a}I$$

$$I\left(1 + \frac{\omega}{a}\right) =$$

Step 4 · Return to the current and find the constant

$$i(t) = \frac{E_0 a \sin(\omega t) - \omega \cos(\omega t)}{L \qquad a^2 + \omega^2} + Ce^{-at}$$

Replace a by R/L and let $D = R^2 + L^2\omega^2$.

$$i(t) = \frac{E_0 [R \sin(\omega t) - L\omega \cos(\omega t)]}{D} + Ce^{-Rt/L}$$

$$i(0) = -\frac{E_0 L\omega}{D} + C = i_0$$

$$C = i_0 + \frac{E_0 L\omega}{D}$$

Problem 18

Each DE in Problems 15–22 is a Bernoulli equation. Solve the given differential equation by using an appropriate substitution.

$$x\frac{dy}{dx} - (1 + x)y = xy^2$$

Step 2 · Use an integrating factor and integrate

On an interval where x has a fixed sign, choose $\mu(x) = xe^x$.

$$\frac{d}{dx}[xe^x u] = -xe^x$$

$$\int xe^x\,dx = xe^x - \int e^x\,dx = (x-1)e^x$$

$$xe^x u = -(x-1)e^x + C = (1-x)e^x + C$$

The usual factor $e^{\int p}$ differs from xe^x only by a constant sign on such an interval, so either factor works.

Step 3 · Return to y and include the zero solution

$$u = \frac{1-x}{x} + \frac{C}{xe^x} = \frac{1-x+Ce^{-x}}{x}$$

$$y(x) = \frac{x}{1-x+Ce^{-x}}$$

Because u = 1/y excluded y = 0, substitute y = 0 into the original equation.

$$y(x) \equiv 0 \quad \text{is also a solution}$$

Keep the denominator nonzero. The derivation assumed $x \neq 0$; any extension through $x = 0$ must be checked in the original equation.

Problem 23

Each DE in Problems 23–30 is of the form given in (5). Solve the given differential equation by using an appropriate substitution.

$$\frac{dy}{dx} = (x + y + 1)^2$$

Step 2 · Integrate and substitute back

$$\int \frac{du}{1+u^2} = \int dx$$

$$\arctan u = x + C \quad \Longrightarrow \quad u = \tan(x + C)$$

$$x + y + 1 = \tan(x + C)$$

$$y(x) = \tan(x + C) - x - 1$$

Step 3 · Check the result and its interval of validity

$$y' = \sec^2(x + C) - 1 = \tan^2(x + C)$$

$$x + y + 1 = \tan(x + C) \quad \Longrightarrow \quad (x + y + 1)^2 = \tan^2(x + C)$$

The left-hand side and right-hand side agree, so the formula solves the original DE.

> Choose an interval that does not cross a pole of tangent. $\cos(x + C)$ must stay nonzero.

Problem 5

A fishery population P(t) with constant harvesting is modeled by

$$\frac{dP}{dt} = P(a - bP) - h, \qquad P(0) = P_0$$

The constants a, b, h, and P_0 are positive. For a = 5, b = 1, and h = 4, show that

$$\frac{dP}{dt} = -(P - 4)(P - 1)$$

Since the DE is autonomous, use the phase portrait concept of Section 2.1 to sketch representative solution curves for $P_0 > 4$, $1 < P_0 < 4$, and $0 < P_0 < 1$. Determine the long-term behavior in each case.

Problem 13

Two chemicals A and B are combined to form a chemical C. The rate or velocity of the reaction is proportional to the product of the instantaneous amounts of A and B not converted to chemical C. Initially there are 40 grams of A and 50 grams of B, and for each gram of B, 2 grams of A are used. It is observed that 10 grams of C are formed in 5 minutes.

How much is formed in 20 minutes? What is the limiting amount of C after a long time? How much of chemicals A and B remains after a long time?

Step 1 · Translate the reaction ratio into amounts

Let X(t) be the grams of C formed after t minutes. Initially, X(0) = 0.

Each 3 grams of product uses 2 grams of A and 1 gram of B.

$$A_{\text{used}} = \frac{2X}{3}, \qquad B_{\text{used}} = \frac{X}{3}$$

$$A_{\text{left}} = 40 - \frac{2X}{3}, \qquad B_{\text{left}} = 50 - \frac{X}{3}$$

The 2:1 ratio describes A consumed : B consumed. It does not mean that forming X grams of C consumes 2X grams of A.

Step 2 · Write the rate equation

The rate is proportional to the product of the unreacted amounts.

$$\frac{dX}{dt} = \kappa\left(40 - \frac{2X}{3}\right)\left(50 - \frac{X}{3}\right)$$

$$\frac{dX}{dt} = \frac{\kappa}{9}(120 - 2X)(150 - X)$$

Set $k = \kappa/9$ to use exactly the constant convention in the TA key.

$$\frac{dX}{dt} = k(120 - 2X)(150 - X) = 2k(60 - X)(150 - X)$$

The physical solution stops before $X = 60$ g while A is still available.